

Induction Motor Load Data

ID	Rated HP	Rated kV	Rated RPM	Qty	PF @ 100%	PF @75%	PF @ 50%	Eff @ 100%	Eff @75%	Eff @ 50%	NLA %	PF @ NL
Comp A	4500	13.2	1800	1	92.83	92.46	89.72	97.99	98.58	99.03	18.24	0.17
Comp B	4500	13.2	1800	1	92.83	92.46	89.72	97.99	98.58	99.03	18.24	0.17
Crusher1	3500	13.2	1800	1	92.83	92.46	89.72	97.99	98.58	99.03	18.24	0.17
Crusher2	3500	13.2	1800	1	92.83	92.46	89.72	97.99	98.58	99.03	18.24	0.17
Comp C	1000	4	1800	1	88.32	86.57	80.57	96.34	97.3	98.06	27.95	0.5
Pump A	450	4	1800	1	89.85	88.26	82.73	94.04	94.86	95.63	23.63	1.02
Pump B	500	4	1800	1	89.85	88.26	82.73	94.04	94.86	95.63	23.63	1.02
Pump C	500	4	1800	1	89.85	88.26	82.73	94.04	94.86	95.63	23.63	1.02

Load Data

ID	Rated kV	Rated kVA	Rated PF	Constant P %	Constant Z %
Distillation Unit	13.8	7000	85	80	20
Receptables A	0.48	400	80	80	20
Receptables B	0.48	350	80	80	20
Light A	0.48	30	83	0	100
Light B	0.48	50	83	0	100
Server Racks A	0.48	35	85	0	100
Server Racks A	0.48	35	85	0	100

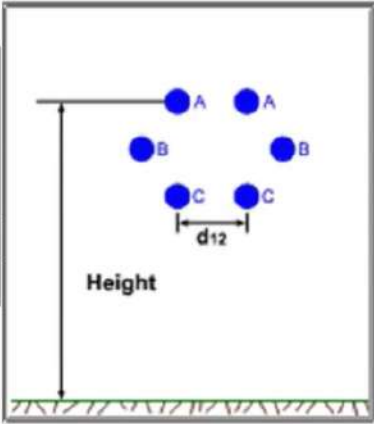
Utility Source Data

ID	Rated kV	LF Mode	MVA _{SC}	X/R	%R +ve	%R -ve	%R zero	%X +ve	%X -ve	%X zero
Power Grid 1	132	Swing	1250	35	0.228	0.228	0.228	7.99	7.99	7.99
Power Grid 2	132	Swing	1600	25	0.249	0.249	0.249	6.24	6.24	6.24

Transmission Line Data

ID	Rated kV	Length km(mile)	Size kcmil	R1 Ohm	X1 Ohm	Y1 uSiemens/km(mile)
Line1	132	12.69(7.89)	397.5	0.117	0.311	22.608(14.048)
Line2	132	9.65(6)	397.5	0.117	0.311	22.608(14.048)

Tower Height 40 ft
Spacing between Phases 5ft
d12 12ft



Transformer Data

ID	Rated MVA	Primary kV	Secondary kV	%Z +ve	X/R	Vector Group
T1	30	132	13.8	9	23.7	Dy1
T2	30	132	13.8	9	23.7	Dy1
T3	2	13.8	4.16	5.75	7.098	Dy1
T4	2	13.8	4.16	5.75	7.098	Dy1
T5	1	4.16	0.48	5.75	5.79	Dy1
T6	1	4.16	0.48	5.75	5.79	Dy1

Cable Data

ID	Rated kV	Length		Size AWG/kcmil	Insulation Type	Cond. Type	#C/Ph	R	X
		m	ft						
C1	138	733	2405	500	EPR	Cu	2	0.035	0.078
C2	138	733	2405	500	EPR	Cu	2	0.035	0.078
C3	15	259	850	1000	EPR	Cu	2	0.022	0.071
C4	15	259	850	1000	EPR	Cu	2	0.022	0.071
Cable1	15	30	100	6	Rubber2	Cu	1	0.51	0.063
Cable2	15	237	777	6	Rubber2	Cu	1	0.51	0.063
EC3	15	30	100	25	XLPE	Cu	1	0.927	0.151
C9	15	401	1316	4	Rubber2	Cu	1	0.321	0.0594
C10	15	401	1316	4	Rubber2	Cu	1	0.321	0.0594
EC10	5	107	350	3/0	EPR	Cu	2	0.081	0.037
EC2	5	137	450	3/0	EPR	Cu	2	0.081	0.037
EC8	5	137	450	3/0	EPR	Cu	2	0.081	0.037
EC6	5	137	450	3/0	EPR	Cu	2	0.081	0.037
EC4	0.6	45	150	10	XLPE	Cu	1	4.265	0.109
EC5	0.6	15	50	10	XLPE	Cu	1	4.265	0.109